

CNC MACHINE MONITORING SYSTEM

Cloud Hosting — Complete Analysis

Scaling from 1 to 100+ customers · customer pricing · serverless (Lambda) feasibility · cost-cutting plan — all decisions in one document.

INTERNAL DOCUMENT — DO NOT SHARE WITH CUSTOMER

1. THE ONE PRICE YOU TELL EVERY CUSTOMER (NEVER CHANGES)

Standard plan (shared cloud): ₹750 per machine per month · minimum ₹5,000 per month per customer · billed annually in advance.

One-time onboarding: ₹2,500 per machine (connection, configuration, testing).

Dedicated server option (only if a customer demands a private server): ₹25,000/month base + ₹500 per machine.

This price works at 1 customer and at 100 customers. Infrastructure upgrades are your internal cost, already covered inside this price at every stage (Section 3 proves it).

2. HOW 100 CUSTOMERS (10 MACHINES EACH) FIT — MULTI-TENANT MODEL

You do **not** create separate servers per customer. All customers share one platform (multi-tenant SaaS) — your software is already built for this: the Company Management module separates every customer's data by company, with their own users, roles and machines. 100 customers × 10 machines = 1,000 machines on one shared platform, upgraded in planned stages below.

3. SCALING STAGES — WHAT YOU UPGRADE, WHEN, WHAT IT COSTS

Stage	Total machines (≈ customers)	Infrastructure	Your cost /month	Cost per machine	Upgrade trigger
Stage 1 (today)	Up to 75 (≈1–8 cust.)	DB m6i.large · MQTT t2.small · API t3.small (current setup)	~₹12,500	₹170–1,250	Machines >75, or DB CPU >70% sustained
Stage 2	75–250 (≈10–25 cust.)	DB m6i.xlarge (16 GB) · MQTT t3.medium · 2× API t3.medium + load balancer	~₹30,000	₹120–400	Machines >250
Stage 3	250–600 (≈25–60 cust.)	DB r6i.xlarge + read replica · 2× MQTT cluster · 3× API + LB · monitoring	~₹70,000	₹115–280	Machines >600
Stage 4	600–1,200 (≈60–120 cust.)	DB cluster + partitioning · MQTT cluster · 4–6 API nodes autoscaling · S3 archive	~₹1,50,000	₹125–250	Machines >1,200 → repeat pattern bigger

Key point: your cost per machine stays ₹115–250 at every stage while the customer pays ₹750. Every upgrade is funded by revenue already collected; each takes 1–2 days with no customer downtime.

4. PROFIT AT EACH MILESTONE (avg. 10 machines/customer, ₹750/machine)

Milestone	Machines	Revenue /month	Infra + ops /month	Profit /month
5 customers	50	₹ 37,500	~₹ 12,500	~₹ 25,000
25 customers	250	₹ 1,87,500	~₹ 35,000	~₹ 1,52,000
60 customers	600	₹ 4,50,000	~₹ 1,10,000 (incl. support engineer)	~₹ 3,40,000
100 customers	1,000	₹ 7,50,000	~₹ 2,50,000 (incl. DevOps + support)	~₹ 5,00,000

At 100 customers: ~₹60 lakh/year profit from hosting alone, plus onboarding income (1,000 machines × ₹2,500 = ₹25 lakh collected along the way).

5. SERVERLESS (LAMBDA) ANALYSIS — WHAT IS POSSIBLE, WHAT IS NOT

5.1 Can each server be replaced by Lambda?

Server	Lambda possible?	Reason
MQTT broker (t2.small)	NO — never	Machines stay connected 24×7 through live TCP connections. Lambda runs for seconds and shuts down — it cannot hold a permanent connection. A broker process must run continuously.
Database (m6i.large)	NO — never	A database must run 24×7 storing data. There is no “Lambda database.” (Aurora Serverless exists, but CNC machines write data day and night, so it never scales down — you pay similar or more, with less control.)
API server (t3.small)	POSSIBLE — but not worth it	Technically yes (Lambda + API Gateway). But this is your cheapest server (₹1,550/month) — maximum saving ~₹1,000/month. Your live dashboards keep WebSocket connections open all day; API Gateway bills per connection-minute, so always-open dashboards can cost more than the small server. Cold starts also make dashboards feel slow.

The trap: your big cost is the database (₹7,000 of ₹12,500). Lambda attacks the ₹1,550 part — the wrong target. Maximum realistic saving from full Lambda migration: ~8% of the bill, with added complexity and slower dashboards.

5.2 Lambda — positives and negatives for your system

Positives (where Lambda helps)	Negatives (why not for the main system)
• Pay only when code runs — idle time is free • Free tier: 1 million runs/month — background jobs cost ~₹0 • No server patching or management • Scales instantly for burst work (report generation) • Perfect fit for: cron jobs, alarm emails, push notification fan-out, PDF/Excel report generation, data cleanup	• Cannot hold 24×7 connections → no MQTT, weak WebSockets • Cold start delay (0.5–2 sec) → dashboards feel slow • Always-open dashboard connections billed per minute → can exceed a flat small server • 15-minute max run time — unfit for continuous data collection • Harder debugging and local testing; architecture rework cost (developer time) far exceeds ₹1,000/month saving

5.3 AWS IoT Core (“serverless MQTT”) — checked and rejected

Option	Cost at 50 machines (1 msg/sec)	Verdict
Your own MQTT broker (t2.small/t3.small)	~₹1,700/month flat — any message volume	KEEP — flat cost, full control
AWS IoT Core (pay per message)	~2.6 million msgs/machine/month → ₹200–250 per machine → ₹10,000–12,000/month	REJECT — 6–7× costlier at your data rate; gets worse as you grow

5.4 Final serverless decision

Keep MQTT + DB + API on servers (as designed). Use Lambda only for background jobs — scheduled crons, alarm/notification sending, report generation. This gives Lambda’s benefits (near-zero cost for spiky work, keeps main servers small as you grow) without touching the parts where serverless fails or costs more.

6. COST-CUTTING PRIORITY ORDER (BIGGEST SAVING FIRST)

#	Action	Saving /month	Risk
1	1-year Reserved Instances (buy after annual customer payment arrives)	~₹4,000 (30–35%)	None — customer already paid yearly
2	Start DB on t3.medium until ~25 machines, then upgrade to m6i.large	~₹4,000 (year one)	None — upgrade is a 1-hour job
3	Run MQTT + API on one t3.medium until ~30 machines	~₹1,500	Low — separate at Stage 2
4	Lambda for cron jobs, notifications, reports	~₹500–1,000	None — and keeps servers small later
	Optimised Stage-1 cost (all four applied)	₹12,500 → ~₹5,500–6,000	>50% cheaper — without Lambda-fying the main system

7. WHAT ELSE CHANGES AS YOU GROW

At this point	You need
~15 customers	Automated monitoring & alerts (Grafana / UptimeRobot — free tools) so you see problems before customers do.
~25 customers	One support person (₹25–35k/month) for onboarding and tickets.
~50 customers	Part-time DevOps (₹40–60k/month), formal SLA (99.5% uptime), automatic data retention (raw 3–6 months, summaries forever).
~100 customers	Full-time DevOps, on-call rotation, disaster recovery in a second AWS region (cost already inside Stage 4).

8. RULES SO YOU NEVER LOSE MONEY

Rule	Why
Minimum ₹5,000/month per customer	Even a 2-machine customer needs onboarding, support and server share — the floor keeps every customer profitable.
Bill annually in advance	No payment chasing across 100 customers; upgrade money is in hand before you need it.
Onboarding fee ₹2,500/machine, never free	Connecting a machine takes real engineer time; the fee also funds Reserved Instance purchases.
Price-review clause in contract	“Prices reviewed annually; changes capped at 10% with 60 days notice.” Legal protection you'll probably never use.

9. WHAT YOU CAN SAFELY PROMISE THE CUSTOMER

Safe to say: “₹750 per machine per month covers cloud hosting, data storage, daily backups, monitoring and support. Same per-machine price at 5 machines or 50 — when you add machines, the platform scales on our side at no extra charge to you.”

Never say: which AWS instances you use, what AWS costs you, or when you upgrade infrastructure — the customer buys a service, not servers.

AWS prices approximate, Mumbai (ap-south-1) region at ₹95/USD, July 2026, including storage, backups and data transfer. IoT Core estimate at 1 message/second/machine. Verify current rates at aws.amazon.com/pricing before contracting.